

Mapping Real Estate in Egypt

DEPI GRAD PROJECT



TEAM 3: DATA WIZARDS

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Project Overview

Data Sources

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Objectives

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Data Cleaning

Data Modeling

Reporting

Project Overview

Mapping Real Estate in Egypt is a data-driven analysis project built to compare investment opportunities in the New Administrative Capital vs New Cairo.

It provides investors and developers with a clear dashboard to visualize price trends, infrastructure quality, and future growth potential, while considering economic factors like inflation and currency devaluation.



Problem Statement

Investors face "**Information Overload**" and conflicting marketing claims. There is a lack of an objective, **data-backed tool** to measure the real Return on Investment while considering economic factors like inflation and currency devaluation.



Data Sources

Data	Source	Format
Dubbizle	https://www.dubbizzle.com.eg/en/new-cairo/ https://www.dubbizzle.com.eg/en/new-capital-city/	CSV/XLSX
PropertyFinder	https://www.propertyfinder.eg/en/search?l=2255&c=1&fu=0&ob=mr https://www.propertyfinder.eg/en/search?l=40470&c=1&fu=0&ob=mr	XLSX
Kaggle	https://www.kaggle.com/datasets/hassankhaled21/egyptian-real-estate-listings https://www.kaggle.com/datasets/mohamedebrahim88/greater-cairo-real-estate-prices-dataset https://www.kaggle.com/datasets/iyadelwy/egypt-housing-prices https://www.kaggle.com/datasets/mohammedaltet/egypt-houses-price	CSV

Data Sources

Data	Source	Format
CAPMAS	https://www.capmas.gov.eg/data/mainSubject/1/subSubject/18	PDF
World Bank Group	https://www.worldbank.org/ext/en/home	PDF
Central Bank of Egypt	https://www.cbe.org.eg/en/	PDF

Questions to answer

- **Market Valuation & Pricing:** What is the specific price disparity between **New Cairo** and the **New Administrative Capital**, and how does the price per square meter reflect the investment value in each?
- **Supply & Inventory Composition:** Which property types dominate the current listings, and how is the inventory distributed between the two cities in terms of volume and unit variety?
- **Delivery & Finishing Readiness:** What is the timeline for property availability (**Ready to Move vs. Future Delivery**), and what level of finishing (**e.g., Super Lux vs. Core & Shell**) characterizes the majority of the current supply?

Project Workflow

steps we're following during our project

Sourcing

Collecting data from
their sources

Cleaning

Fixing problems with the data
coming from their source

Modeling

Making relations between
different tables using STAR
schema

DAX

Performing DAX calculations
using formulas

Sample Visual

Explaining what some
visuals are used for

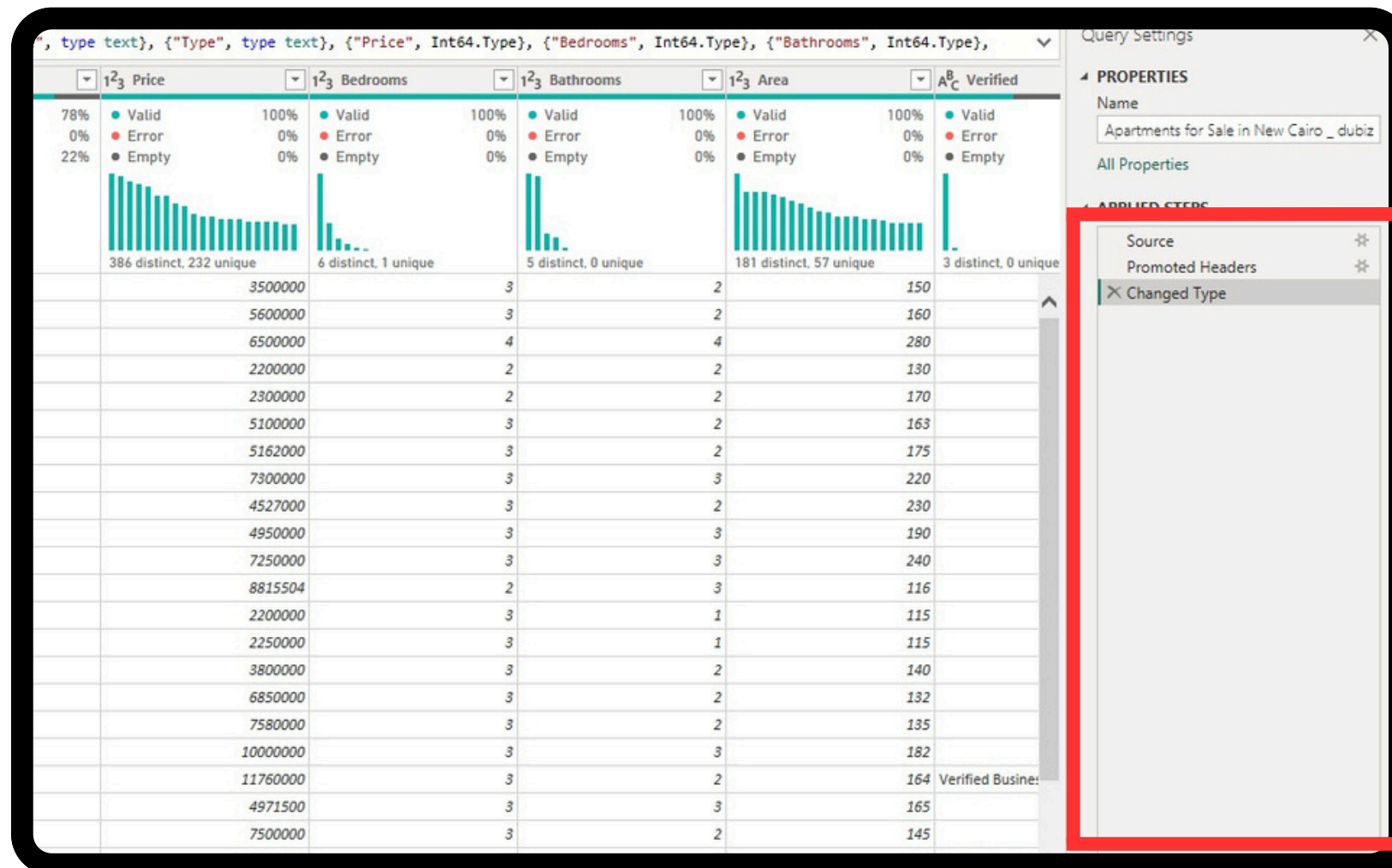
Reports + Dashboards

Delivering 5 Reports +
1 Big Summary Dashboard

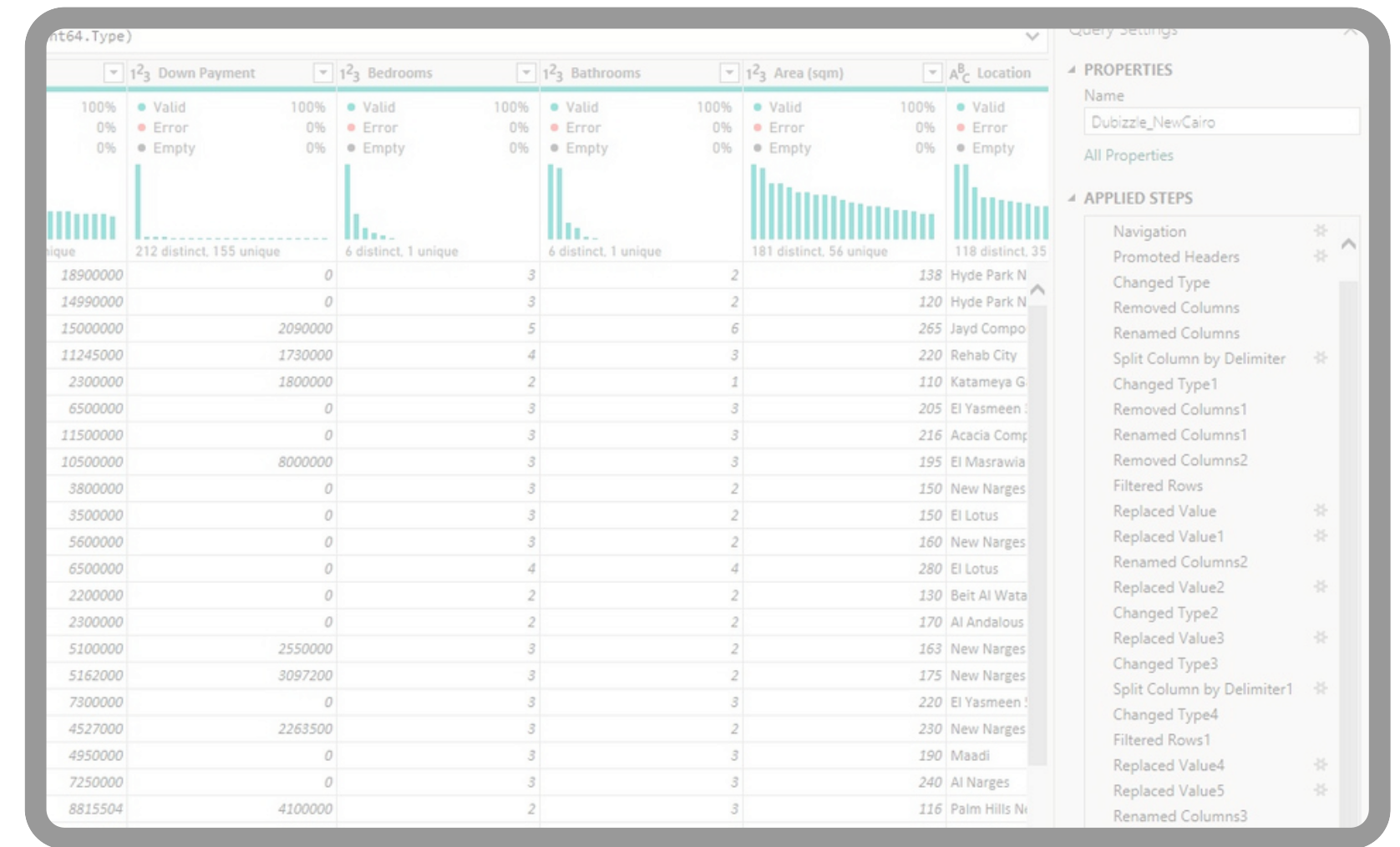
Data Cleaning

Refining the dataset and using "Data Imputation" techniques to handle inconsistent or missing information.

BEFORE



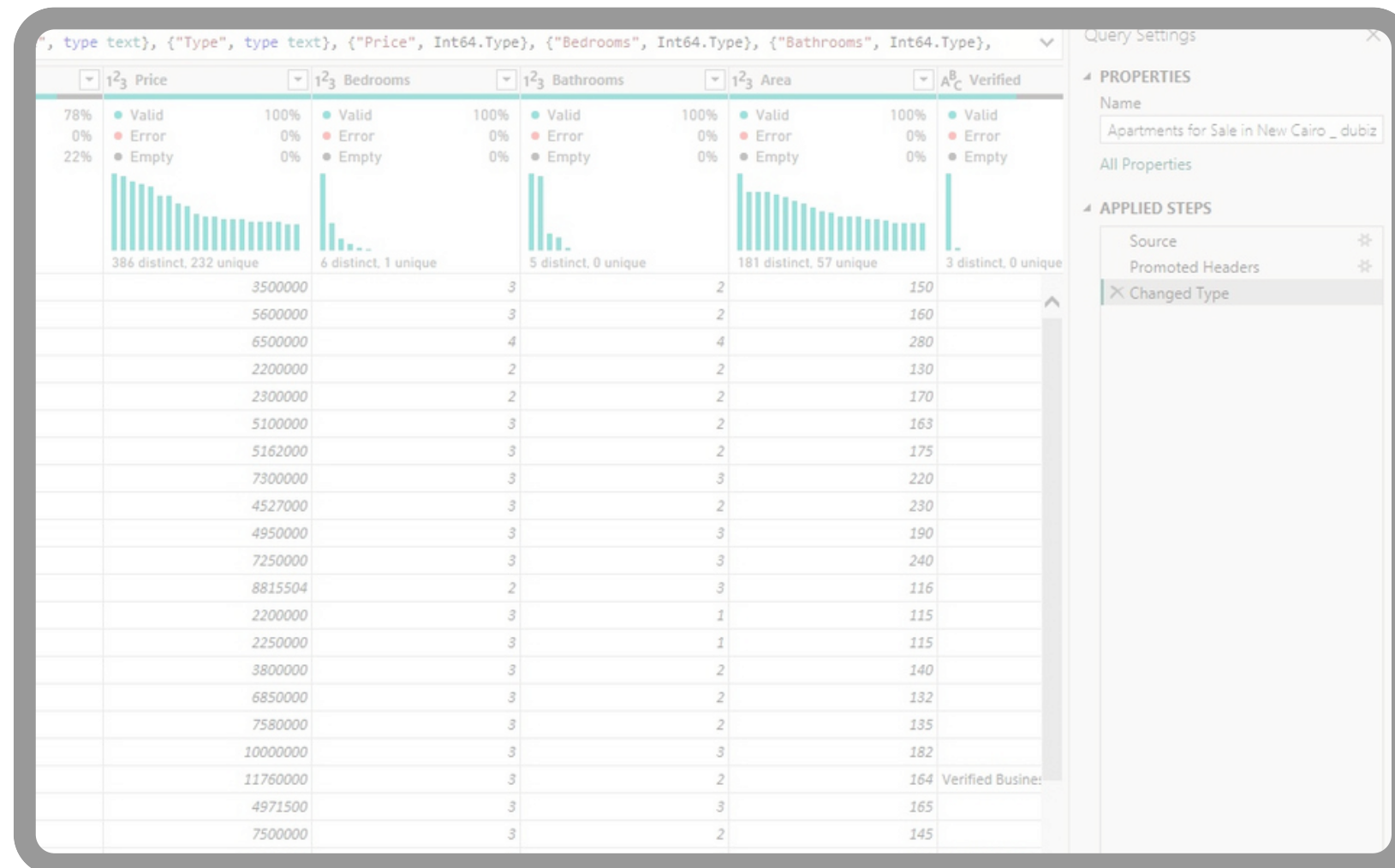
AFTER



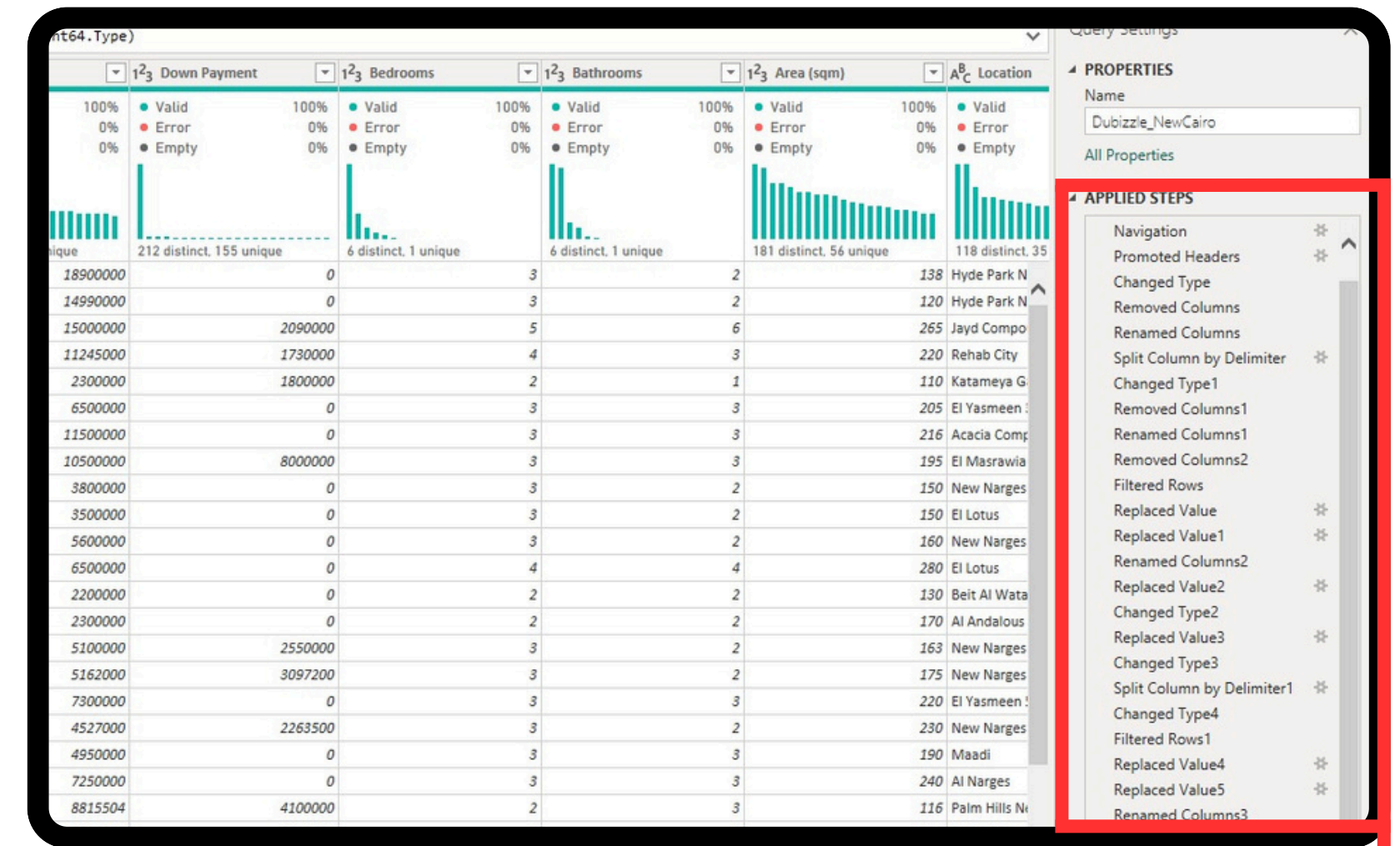
Data Cleaning

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BEFORE



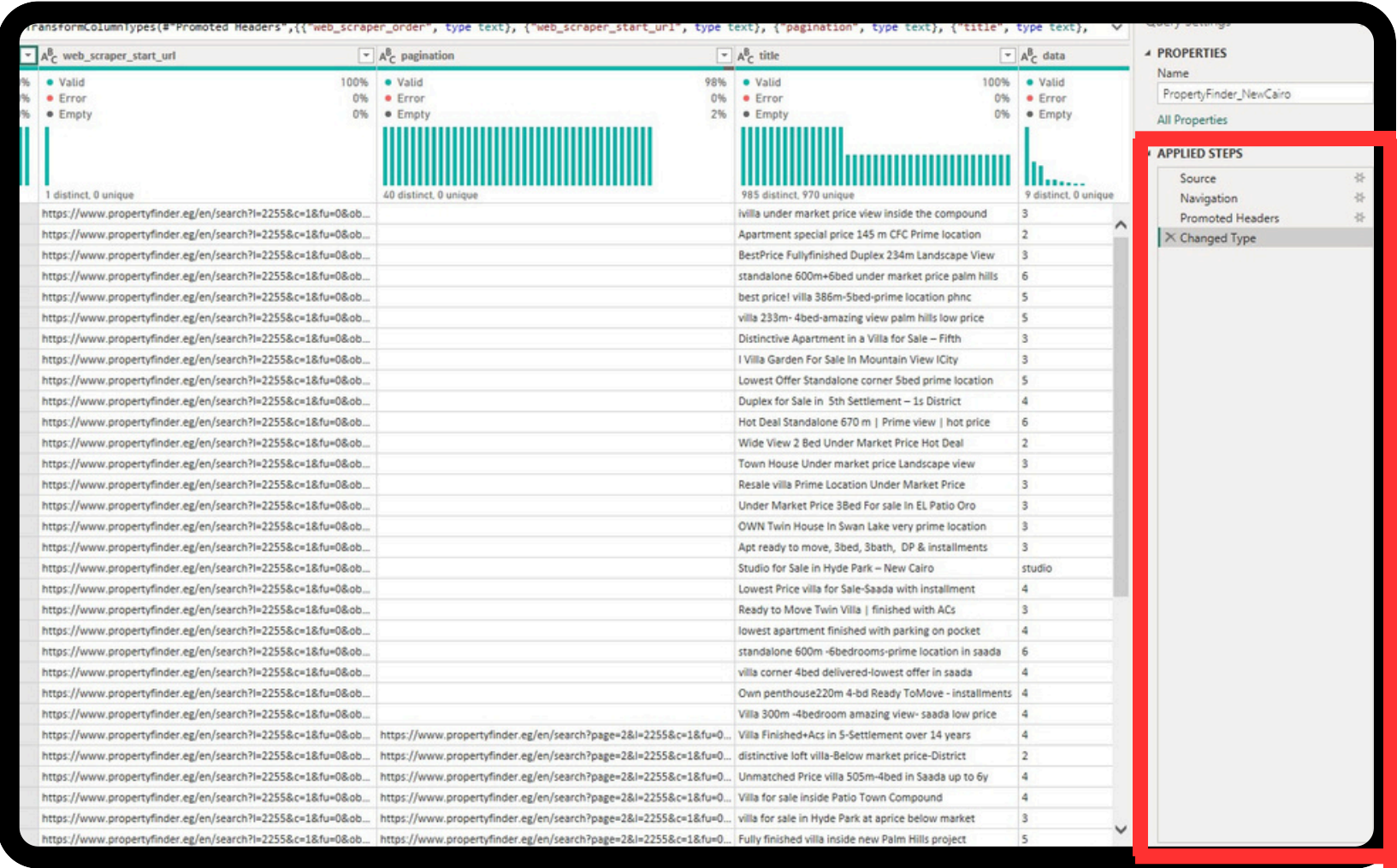
AFTER



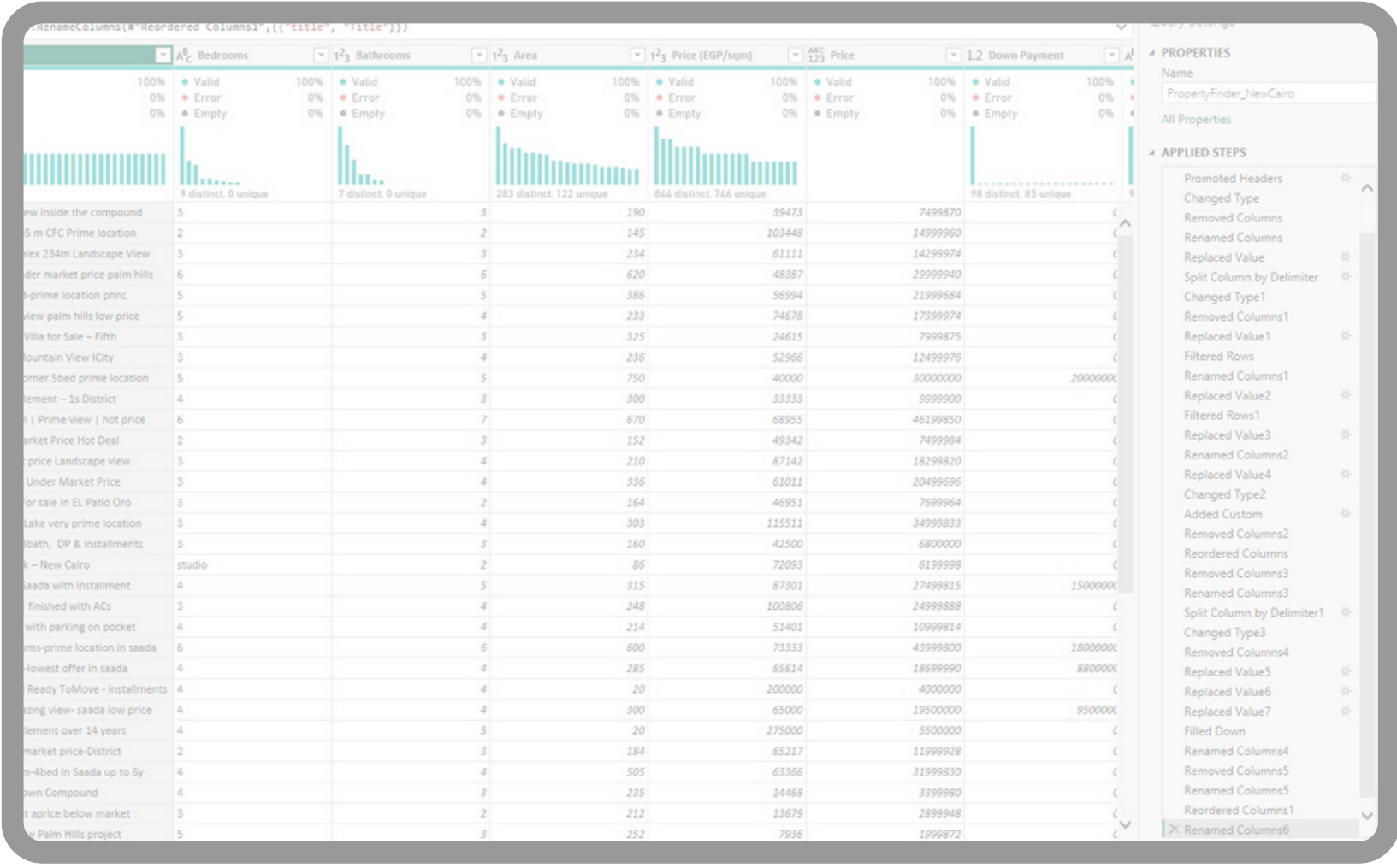
Data Cleaning

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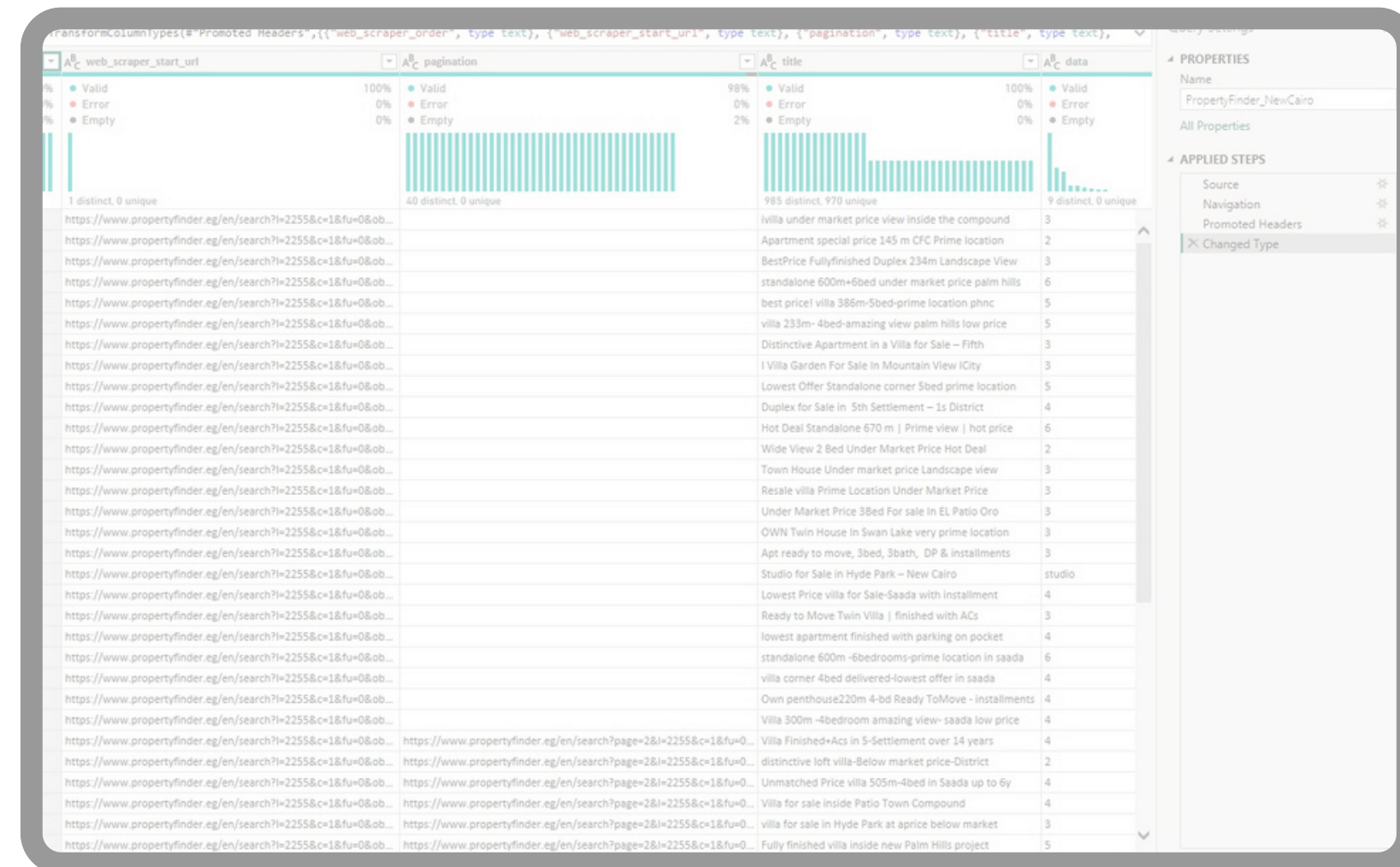
AFTER



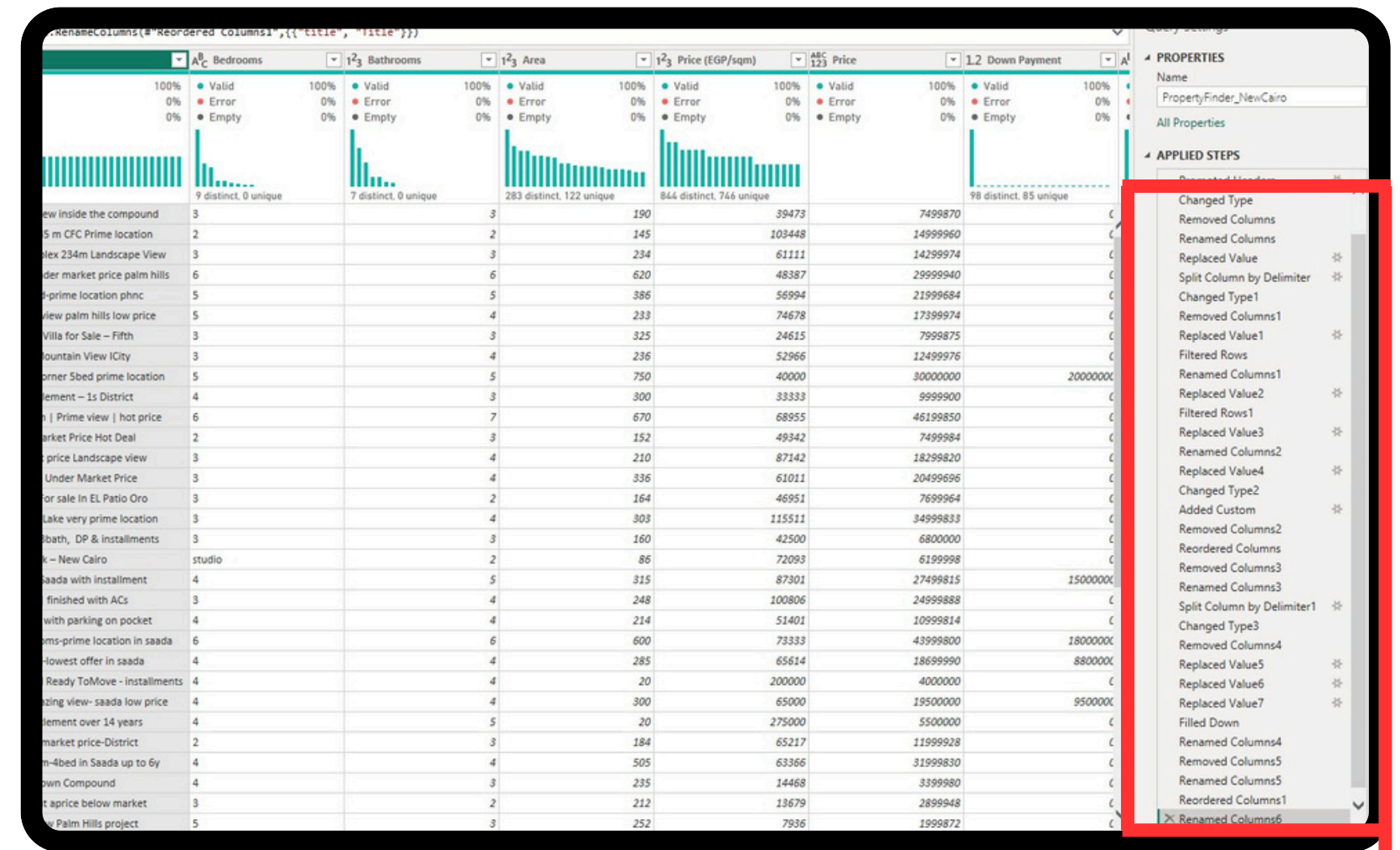
Data Cleaning

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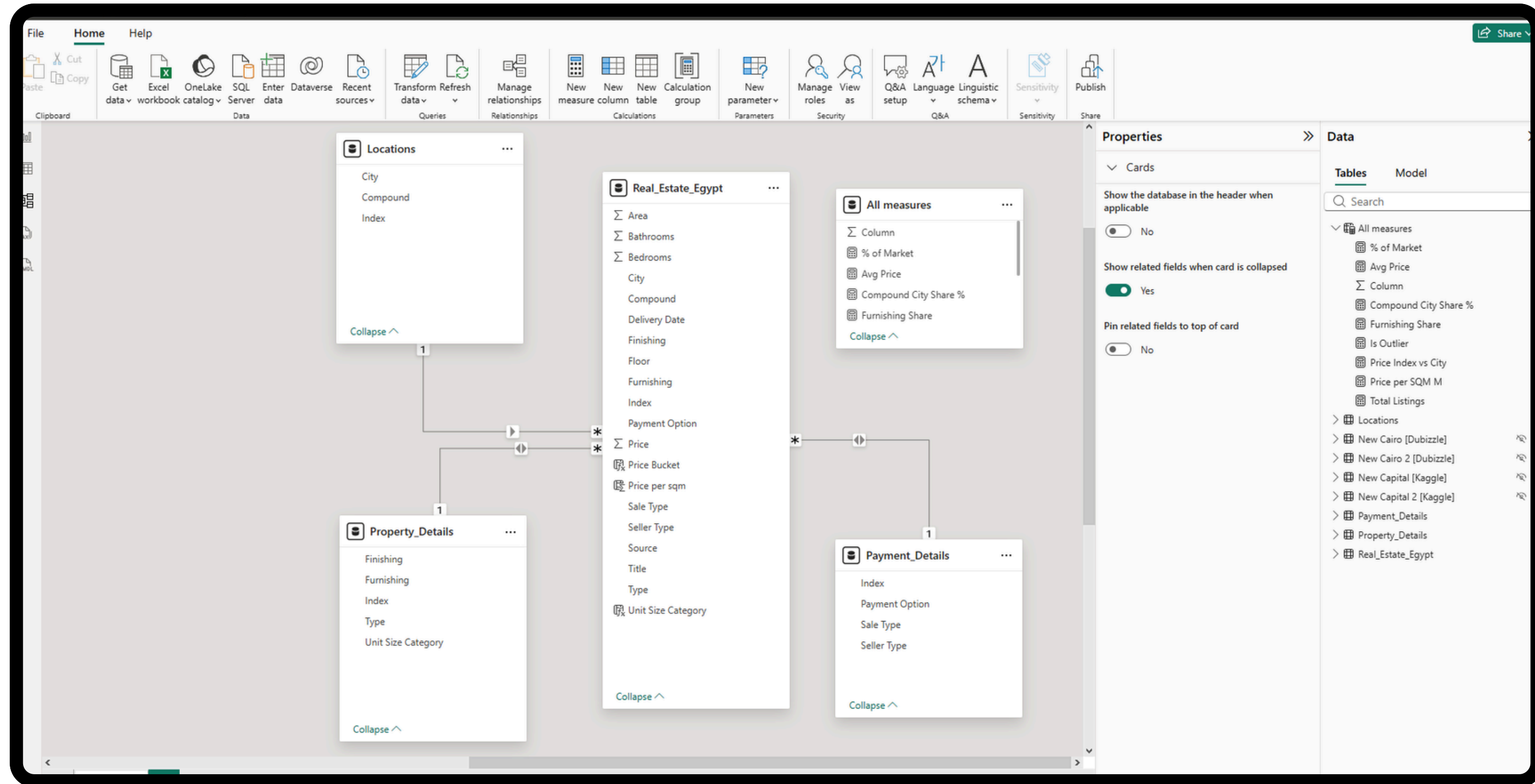
BEFORE



AFTER



Data Modeling



DAX

A formula language in Power BI used to create custom calculations, measures, and calculated columns for advanced data analysis, modeling, and dynamic reporting.

```
1 % of Market =  
2 DIVIDE(  
3     [Total Listings],  
4     CALCULATE([Total Listings], ALL('Real_Estate_Egypt'))  
5 )
```

```
1 Avg Price = COALESCE(AVERAGE('Real_Estate_Egypt'[Price]), 0)
```

```
1 Price Index vs City =  
2 VAR AvgCityPrice =  
3     CALCULATE(  
4         [Price per SQM M],  
5         ALLEXCEPT('Real_Estate_Egypt', 'Real_Estate_Egypt'[City])  
6     )  
7 VAR CurrentPrice = [Price per SQM M]  
8 RETURN  
9 DIVIDE(CurrentPrice - AvgCityPrice, AvgCityPrice)
```

```
1 Is Outlier =  
2 VAR Q1 = PERCENTILEX.INC(ALLSELECTED('Real_Estate_Egypt'), 'Real_Estate_Egypt'[Price], 0.25)  
3 VAR Q3 = PERCENTILEX.INC(ALLSELECTED('Real_Estate_Egypt'), 'Real_Estate_Egypt'[Price], 0.75)  
4 VAR IQR = Q3 - Q1  
5 VAR UpperBound = Q3 + (1.5 * IQR)  
6 VAR LowerBound = Q1 - (1.5 * IQR)  
7 VAR CurrentPrice = SELECTEDVALUE('Real_Estate_Egypt'[Price])  
8 RETURN  
9 IF(CurrentPrice > UpperBound || CurrentPrice < LowerBound, 1, 0)
```

Data

Search

▼ All measures

- ☐ % of Market
- ☐ Avg Price
- ☐ Σ Column
- ☐ Compound City...
- ☐ Furnishing Share
- ☐ Is Outlier
- ☐ Price Index vs C...
- ☐ Price per SQM M
- ☐ Total Listings

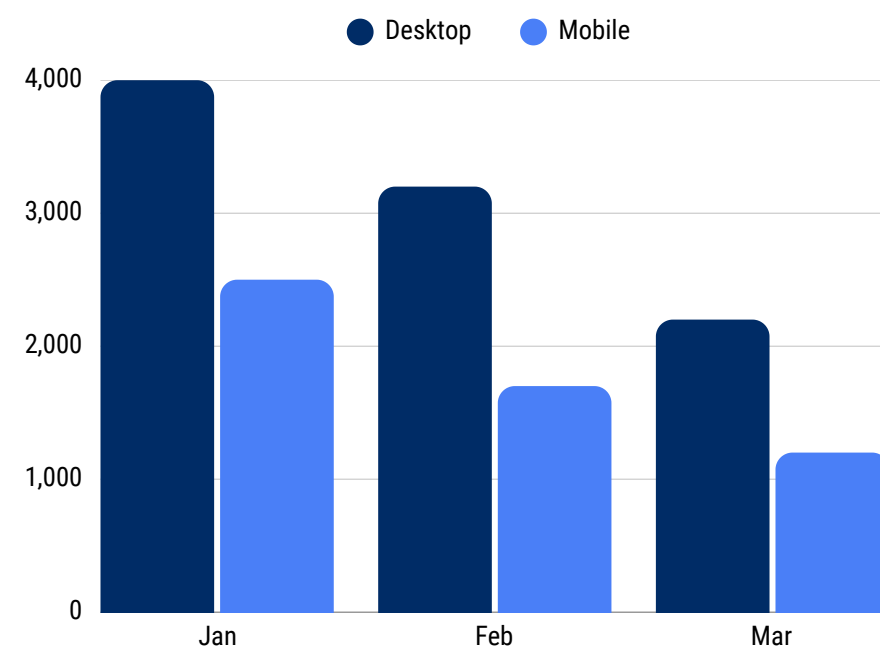
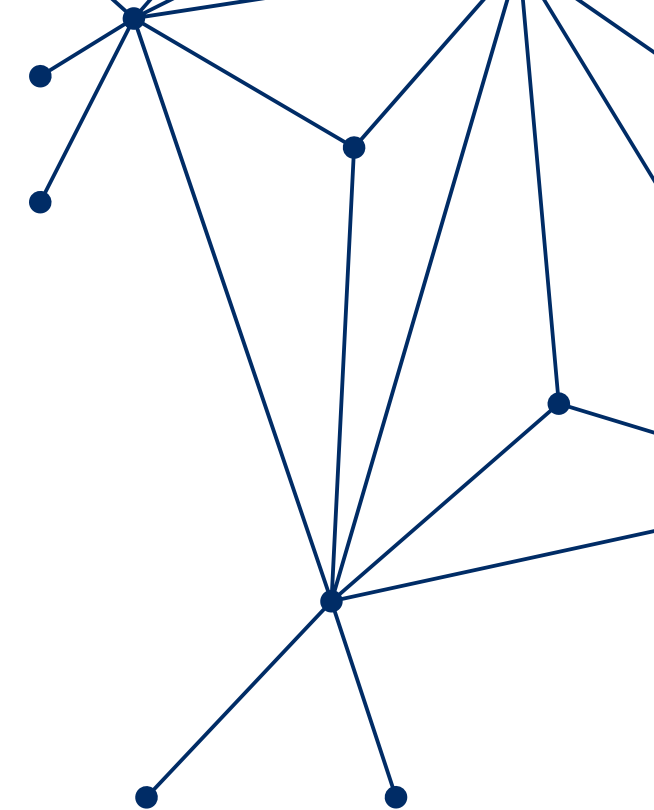
> Locations

> Payment_Details

> Property_Details

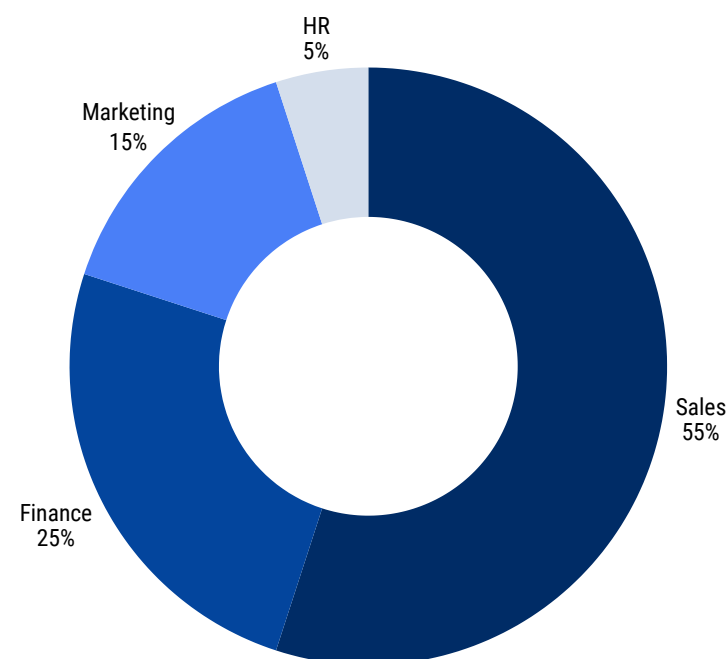
> Real_Estate_Egypt

Sample Visual



Clustered Bar Chart

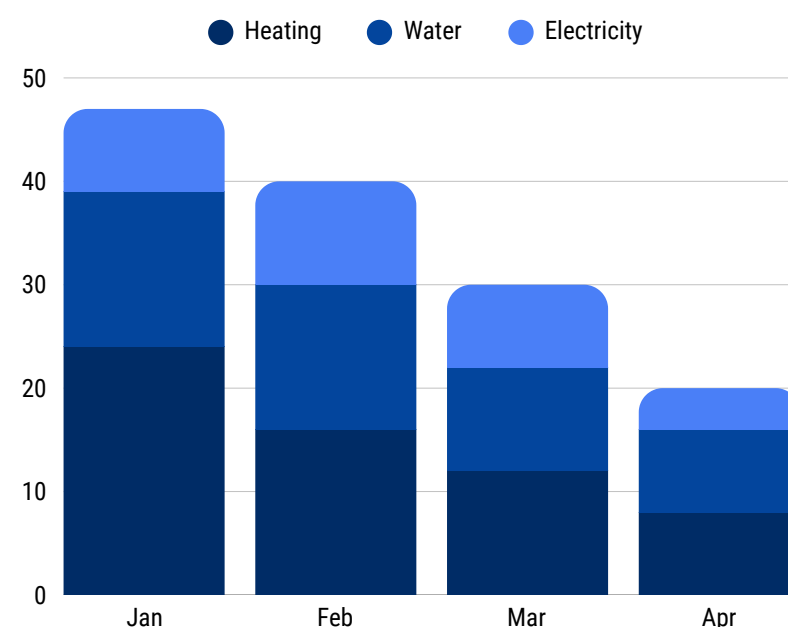
These charts are used to perform side-by-side comparisons between multiple data series **across distinct categories**. They are designed to highlight differences and variances between groups, making it easy to identify trends and compare individual values directly within a single view.



Donut Chart

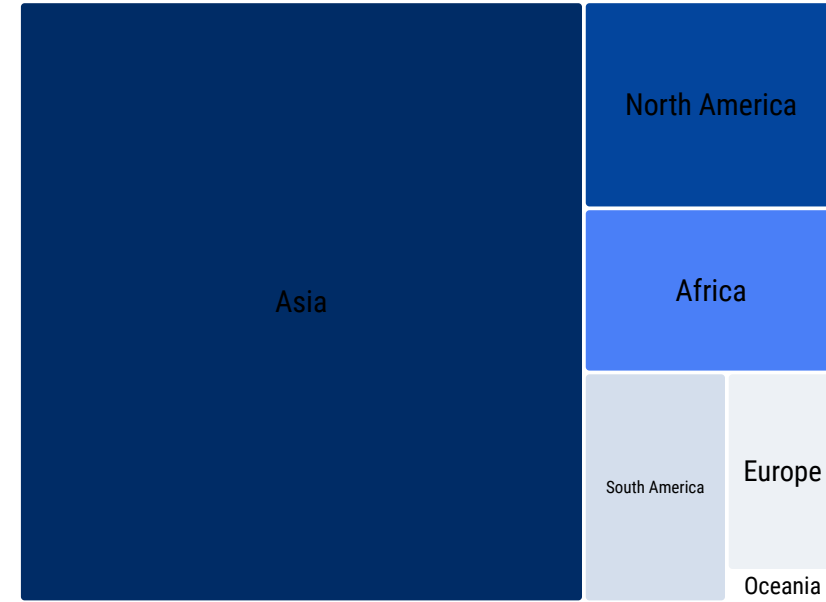
These visuals are ideal for representing **"part-to-whole" relationships and illustrating categorical proportions**. By focusing on the arc length, they allow the viewer to quickly understand the distribution of a total amount among various segments and evaluate the relative weight of each category.

Sample Visual



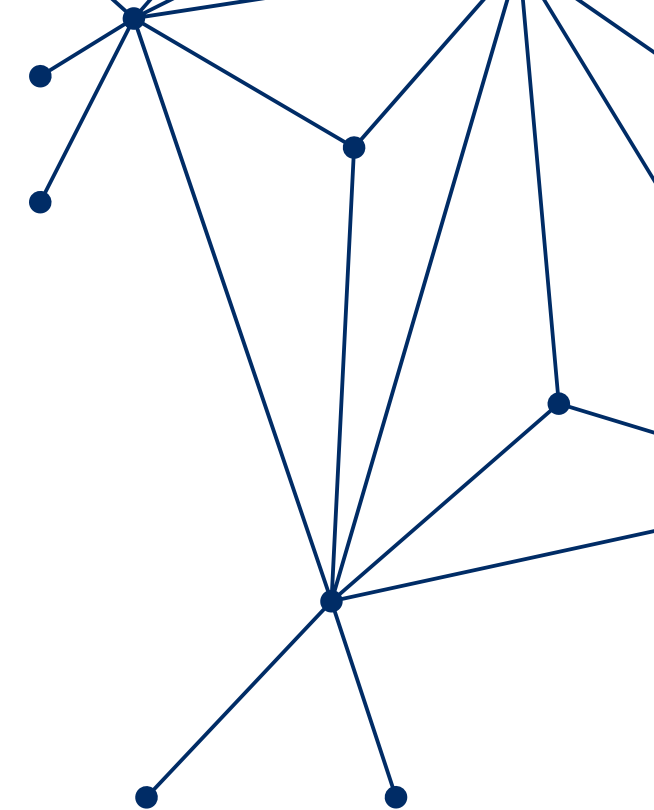
Stacked Column Charts

These charts are used to visualize the internal composition of a total value by breaking it down into **sub-categories**. They provide a clear view of how different components contribute to the overall sum, while still allowing for a primary comparison of total volumes across different data points.

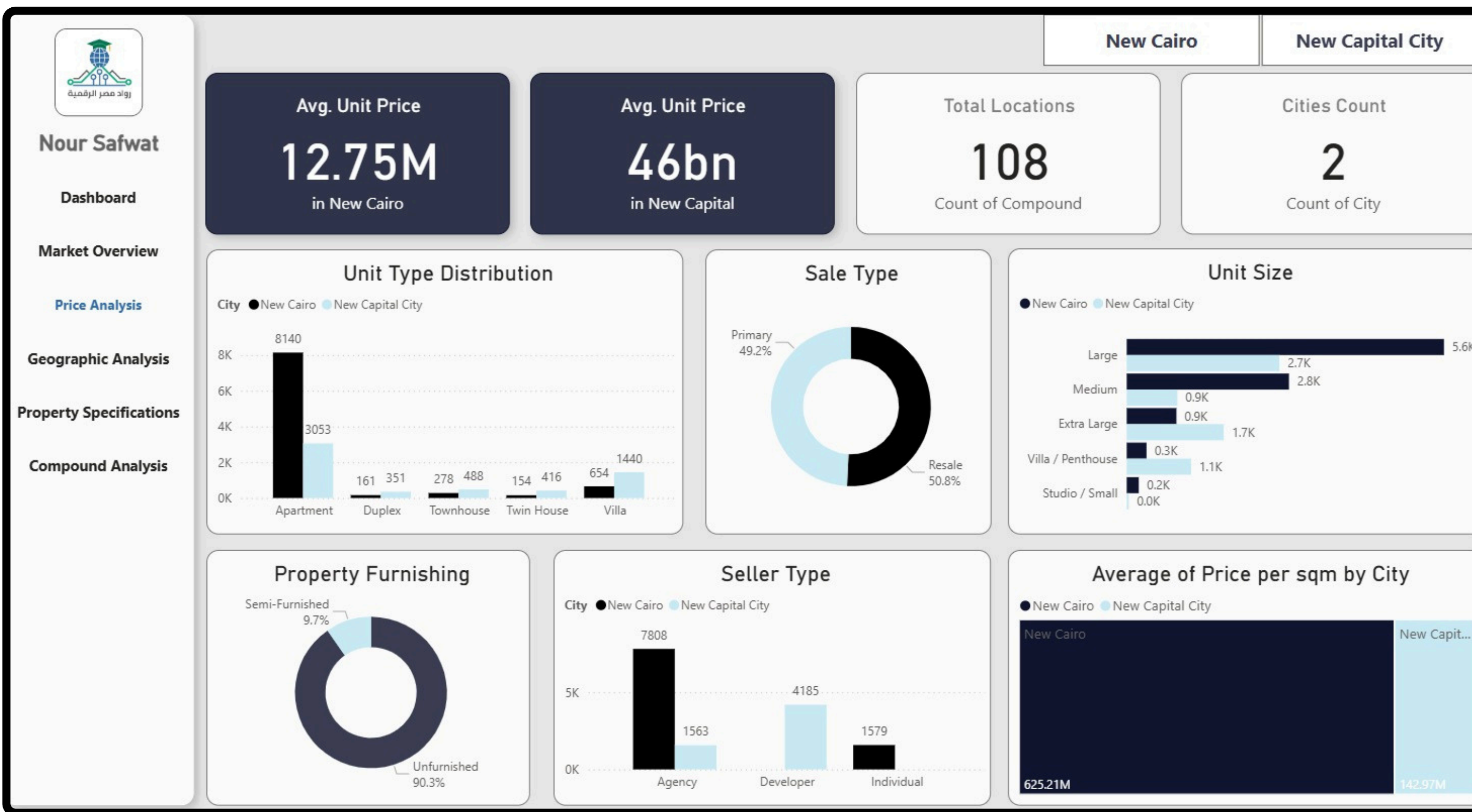


Treemaps

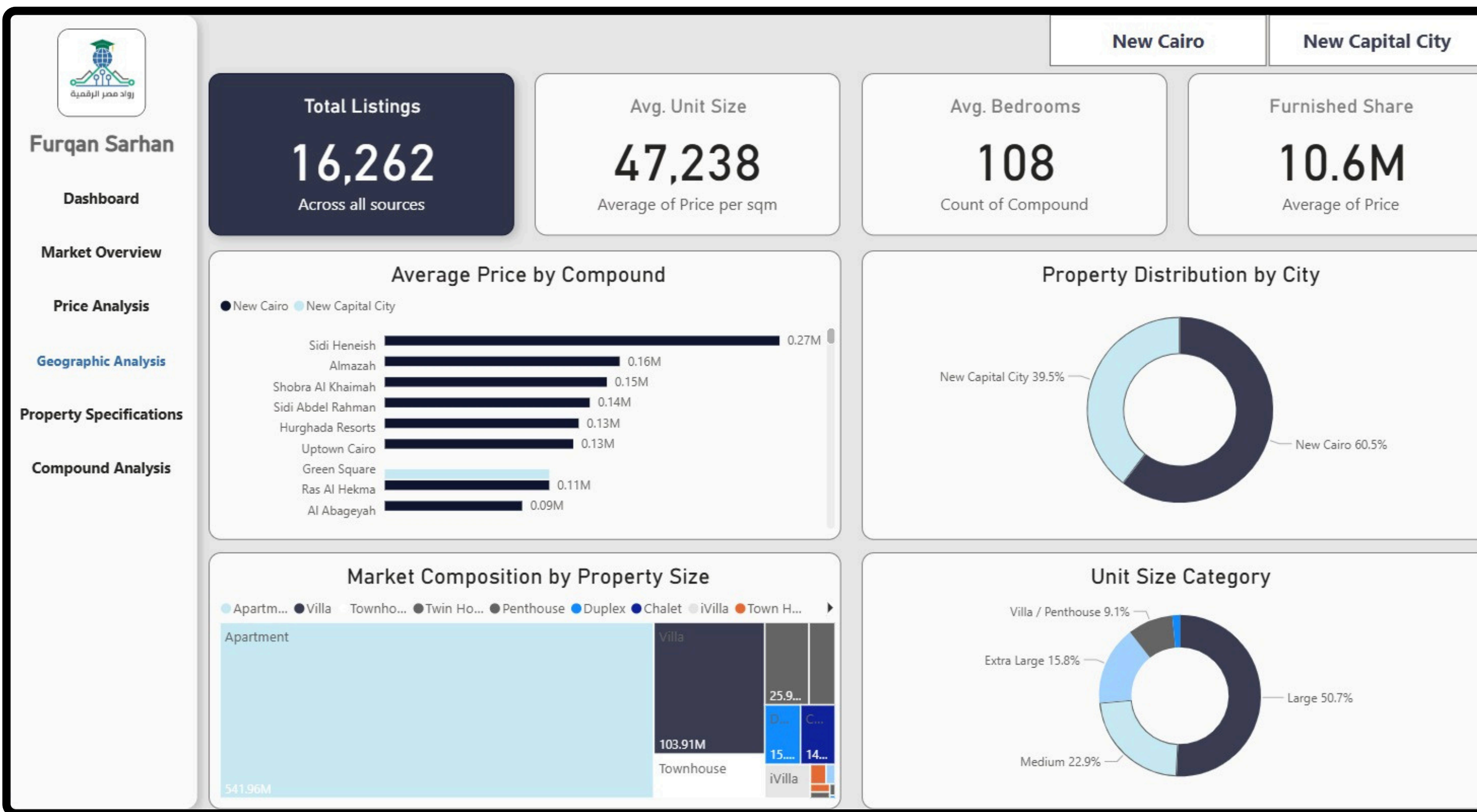
Treemaps are used to display large sets of hierarchical data through nested rectangles of varying sizes. This visual is effective for **ranking categories and identifying dominant patterns** within complex datasets, where the area of each rectangle is proportional to the value it represents.



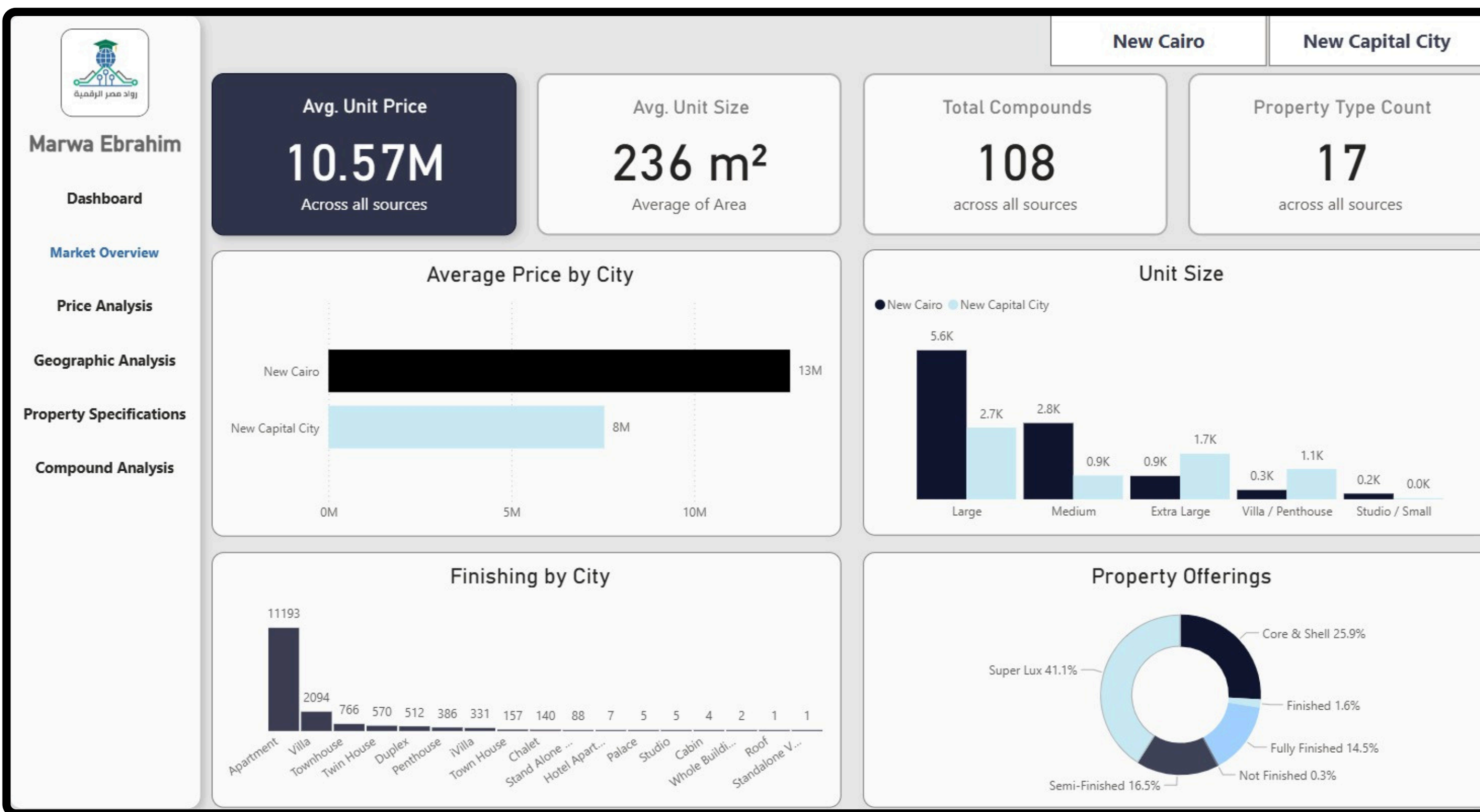
Report 1: Price Analysis



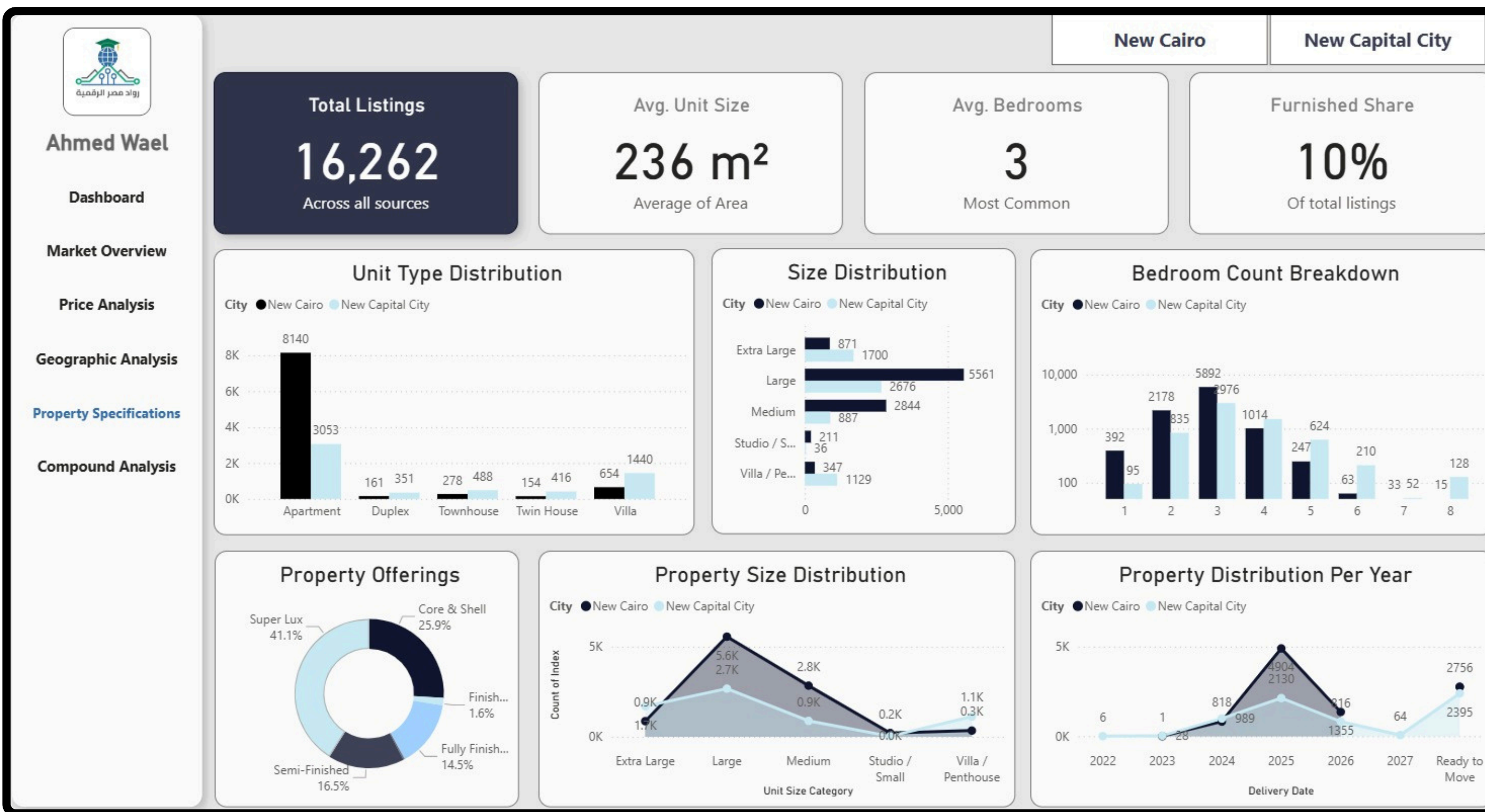
Report 2: Geographic analysis



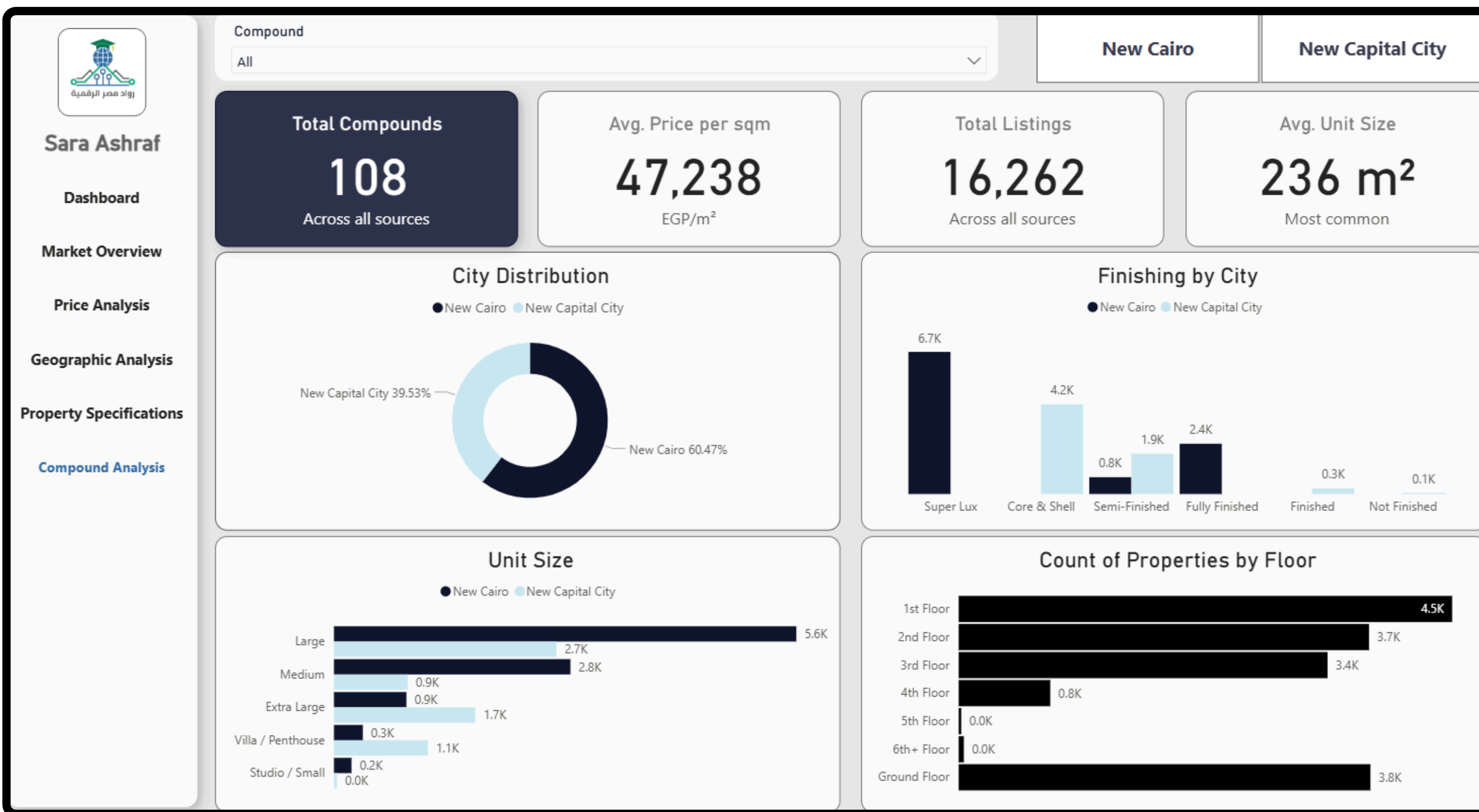
Report 3: Market Overview



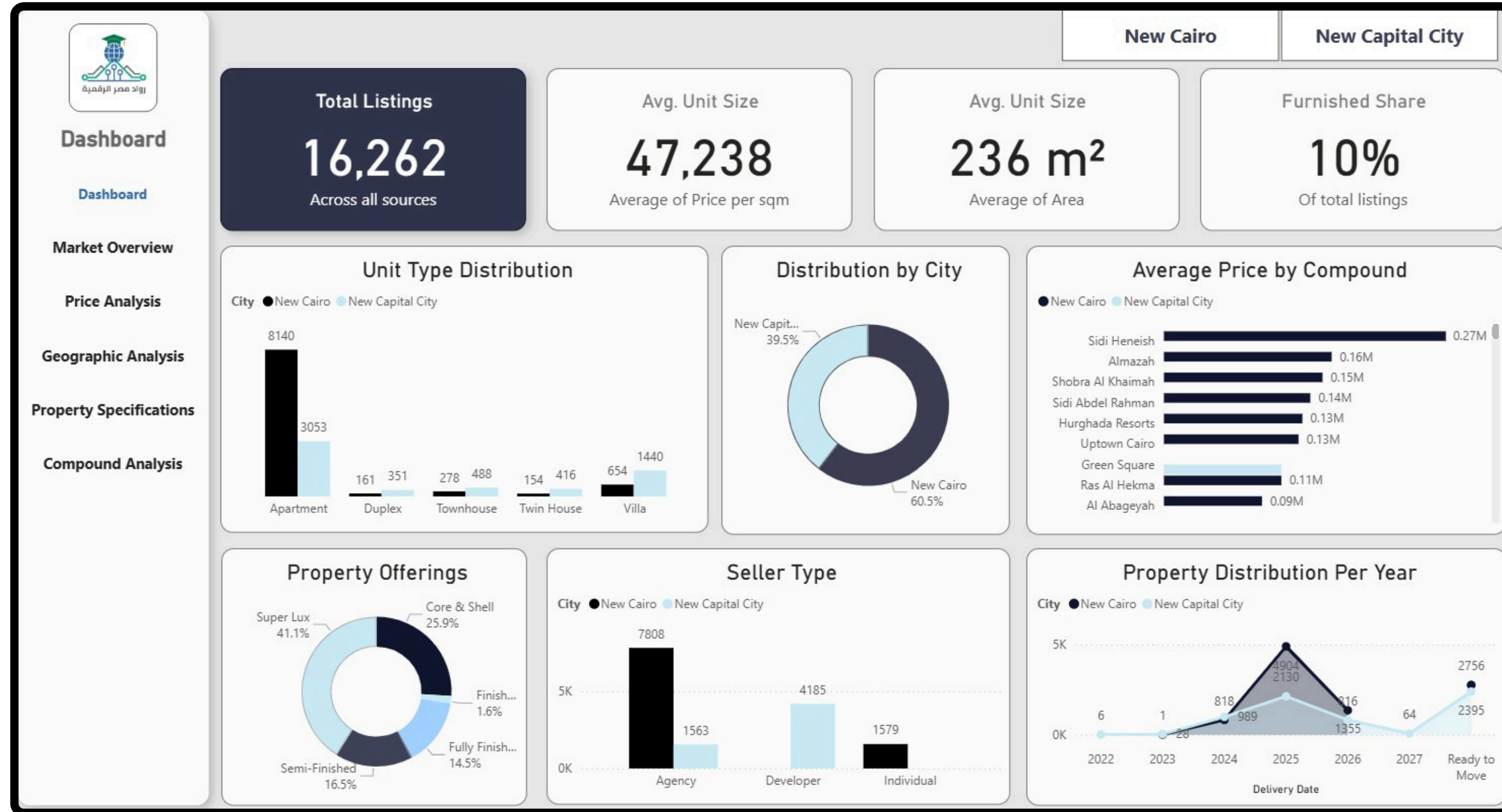
Report 4: Property specifications



Report 5: Compound analysis



Dashboard



Recommendations & Insights

- The Egyptian real estate market is large, dynamic, and highly active, yet it follows clear and consistent patterns rather than randomness.
- Market demand is primarily driven by families, as reflected in the dominance of medium-to-large units and 3-bedroom properties.
- Pricing is not fixed; it is shaped by the interaction of multiple factors, including location, property specifications, and market supply.
- The continuous increase in listings over time reflects a rapidly expanding and evolving real estate market.
- The dominance of unfurnished units highlights buyers' preference for flexibility and customization.

Ultimately, successful real estate decisions are not based on a single metric, but on a comprehensive understanding of the entire market landscape.

